

1 Summary statistics for treatment efficacy estimation in clinical
2 trials: when simplicity rivals complexity

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5 **Contents**

6	1 Introduction	2
7	1.1 The summary measures tradition	3
8	1.2 Extensions and refinements	4
9	1.3 MMRM: strengths and practical limitations	4
10	1.4 When are summary statistics adequate?	5
11	1.5 Present study	6
12	2 Methods	6
13	2.1 ADEMP structure	6
14	2.2 Data generating model	7
15	2.3 Parameter values	7
16	2.4 Design specifications	7
17	2.5 Missing data mechanism	8
18	2.6 Simulation procedure	9
19	2.7 Performance metrics	9
20	3 Results	9
21	3.1 Headline findings	9
22	3.2 Type I error	9
23	3.3 Summary statistics	10
24	3.4 Figures	11
25	4 Discussion	11
26	5 Future research	13
27	6 Appendix: Morris et al. (2019) ADEMP compliance audit	14
28	References	15

29 **Abstract**

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Mixed-model repeated measures (MMRM) is the default primary analysis for longitudinal continuous endpoints in clinical trials, but the two-stage summary-measures approach (per-subject slope or change score, followed by ANCOVA) remains attractive where covariance misspecification, degrees-of-freedom controversy, and interpretability are practical concerns. We synthesize three decades of summary-measures literature and report a Monte Carlo simulation, under a random-intercept-and- slope generating model, comparing slope-summary ANCOVA, change-score ANCOVA, and MMRM on bias, coverage, power, and Type I error, each scored against its own estimand and each reported with Morris-Table-6 Monte Carlo standard errors. We argue that the most defensible framing for summary statistics is not a claim of general superiority over MMRM, which the literature (Frison and Pocock; Vossoughi; Ard) has already established for the linear-divergence setting the current simulation covers, but a decision-support case grounded in the ICH E9(R1) estimands framework: the summary-statistic slope estimand is covariance-model-free and directly interpretable, while the MMRM visit-specific estimand is conditional on the working covariance model.

Keywords: summary measures; ANCOVA; MMRM; longitudinal clinical trials; estimands; Monte Carlo simulation

Data and code availability

All simulation code, the manuscript source, and the derived results object discussed here are version-controlled in the `summarystats` repository (`analysis/scripts/sim_study.R`, `analysis/report/report.Rmd`, `analysis/data/derived_data/sim_09.rds`). No human subjects data are used; all data are simulated. See the Reproducibility note at the end of the Results section for the exact command to regenerate the simulation output.

1 Introduction

Mixed-model repeated measures (MMRM) has become the default analytic strategy for longitudinal clinical trials with continuous endpoints. Regulatory agencies and methodological guidance documents increasingly endorse MMRM as the primary analysis¹, and the approach has clear theoretical advantages: it uses all available data under the missing-at-random (MAR) assumption, accommodates unbalanced designs, and models complex within-subject correlation structures².

Yet MMRM carries substantial practical burdens. The method requires specification of a covariance structure, and misspecification can inflate Type I error or reduce power³. Degrees-of-freedom approximations (Satterthwaite versus Kenward-Roger) remain unsettled, with different software implementations yielding discrepant p-values for the same data^{4,5}. Convergence failures are common in R packages when fitting unstructured covariance matrices, particularly with small samples or many time points. And perhaps most critically, the MMRM treatment effect estimate at a single visit is not always intuitive to clinical investigators, who may prefer a single, interpretable summary of each patient's response trajectory.

68 An older and simpler tradition, the summary measures approach, offers a compelling alter-
69 native. First articulated systematically by⁶, this two-stage method reduces each subject's
70 repeated measurements to a single scalar summary (e.g., a slope, area under the curve, post-
71 treatment mean, or change score) and then applies standard between-group tests (t-tests,
72 ANCOVA) to these summaries. The approach requires no covariance structure specification,
73 yields exact p-values from well-understood distributions, and produces treatment effect esti-
74 mates that are directly interpretable.

75 We ask whether there exist clinical trial settings in which summary statistic-based inference
76 is as effective as or superior to MMRM, and what those settings look like.

77 1.1 The summary measures tradition

78 The intellectual history of summary measures in clinical trials spans three decades.⁶ introduced
79 the two-stage framework in the BMJ, arguing that the then-common practice of testing group
80 differences at each time point was both statistically invalid (due to multiplicity) and clinically
81 uninformative. They proposed that investigators first identify a summary that captures the
82 clinically relevant feature of each patient's trajectory, then analyze these summaries with
83 simple methods. The approach is statistically valid because the summaries, being one per
84 subject, are independent across subjects.

85 ⁷ extended this framework substantially by demonstrating that when the data can be summa-
86 rized by pre-treatment and post-treatment means, analysis of covariance (ANCOVA) of the
87 post-treatment means adjusted for the pre-treatment means is the method of choice. They
88 quantified the superiority of ANCOVA over analysis of post-treatment means alone and over
89 analysis of change scores, showing that ANCOVA dominates both in terms of reduced variance
90 and avoidance of bias. Importantly, they showed that having more than one pre-treatment
91 measurement provides meaningful gains in precision, because the pre-treatment mean is a
92 better estimate of the true baseline than a single observation.

93 ⁸ addressed the case where treatment effects diverge linearly over time. For this setting, the
94 natural summary statistic is each subject's regression slope, and the authors showed that
95 adjusting slopes for baseline values (or better, for estimated intercepts) yields substantial
96 efficiency gains. They derived the optimal linear summary statistic for any repeated measures
97 design with known covariance structure and known shape of the mean treatment difference,
98 demonstrating that the summary measures approach can be made nearly fully efficient.

99 ⁹ provided a practical synthesis, arguing that much repeated measures analysis using mixed
100 models could be replaced by a summary measures approach with negligible loss of efficiency.
101 Senn and colleagues described strategies for baseline adjustment with multiple baselines and
102 proposed a compromise trend/mean measure (regression through the origin) useful when both
103 the level and rate of change matter. They also made the provocative observation that some
104 repeated measures approaches implicitly violate intention-to-treat principles, a problem that
105 summary measures avoid.

1.2 Extensions and refinements

¹⁰ recognized that summary statistic distributions depend on the pattern of missingness and proposed stratifying summary statistic tests by missing data pattern. Simulations showed that stratification increases power and improves robustness to violations of missing data assumptions.¹¹ later provided a unified sample size framework for summary statistics, covering slopes, post-baseline means, change scores, and final observations, with modifications for missing data from staggered entry and random dropouts.

¹² examined the efficiency of rank and permutation tests based on summary statistics under more complex distributional assumptions, including censored laboratory markers. Their work demonstrated that the choice of summary statistic can substantially affect asymptotic relative efficiency, particularly when the repeated measures exhibit nonlinear growth or left censoring.

¹³ revisited the choice between ANCOVA and change-score analysis, clarifying persistent confusion in the literature. Senn argued that ANCOVA is generally preferable for randomized trials because it adjusts for chance imbalances in baseline values without introducing the bias that afflicts change-score analysis in non-randomized settings.

1.3 MMRM: strengths and practical limitations

The theoretical foundation for MMRM rests on the random-effects model of², developed further in the two-stage random-effects lineage summarized by¹⁴, which provides maximum likelihood and restricted maximum likelihood (REML) estimation for models with structured covariance. The key practical advantage is that MMRM uses all available data, providing valid inference under MAR without requiring imputation. The DIA/PhRMA working group formalized MMRM as the recommended primary analysis for continuous longitudinal endpoints¹⁵, and¹⁶ compared MMRM against LOCF-ANCOVA across 25 New Drug Application datasets, reinforcing the regulatory lineage that led to MMRM's current default status.

¹⁷ provided an influential empirical comparison of MMRM and LOCF-ANCOVA across eight duloxetine trials. While MMRM and LOCF agreed on significance in 82% of 202 comparisons, MMRM yielded significant results in 12.4% of cases where LOCF did not. This work cemented MMRM's position as the preferred analysis but also revealed that the two approaches diverge substantively in a non-trivial minority of cases.

However, MMRM faces several practical limitations that merit mention:

1. **Degrees-of-freedom approximations.** @kenwardSmallSampleInference1997 proposed a scaled Wald statistic with an F approximation that reduces small-sample bias, but the Kenward-Roger adjustment depends on the parameterization of the covariance model, producing different results across software implementations. The alternative Satterthwaite approximation is computationally simpler but can be liberal in small samples. Neither method has a clear theoretical advantage in all settings.
2. **Software fragmentation.** SAS PROC MIXED has long been the reference implementation, but R packages (`nlme`, `lme4/lmerTest`, `glmmTMB`) have exhibited convergence

144 problems, particularly with unstructured covariance matrices. The `mmrm` package [mm-
145 rmPackage2024] was developed specifically to address these gaps, but discrepancies be-
146 tween SAS and R results persist due to different parameterizations.

147 **3. Covariance structure specification.** @vossoughiSummaryMeasureAnalysis2012
148 showed that misspecification of the error covariance structure in LMM can produce
149 non-sensible results, whereas the summary measure approach is robust to the true
150 covariance structure. In practice, analysts often default to unstructured covariance, but
151 this requires estimation of $p(p + 1)/2$ parameters for p time points, which may exceed
152 the information in the data.

153 **4. Interpretability.** The MMRM treatment effect at a specific visit (e.g., week 24) is a
154 conditional quantity that depends on the covariance model. The analogous summary
155 statistic (e.g., slope difference) has a direct clinical interpretation: how much faster does
156 the treatment group improve per unit time?

157 **5. Time-by-covariate interactions.** @liuMixedModelsCovariateInteractions2021
158 demonstrated that MMRM must include time-by-covariate interactions to achieve
159 power gains and robustness against dropout bias relative to complete-case ANCOVA.
160 Without these interactions, MMRM may not outperform a simple ANCOVA of the
161 final visit.

162 1.4 When are summary statistics adequate?

163 The performance comparison literature provides guidance on when summary statistics match
164 or exceed MMRM.³ conducted extensive Monte Carlo simulations comparing the summary
165 measure approach (SMA), linear mixed models (LMM), and the unstructured multivariate
166 approach (UMA) for linear trend data. They found that SMA completely dominated
167 UMA and performed convincingly close'' to the best-fitting LMM in testing all
168 effects. Critically, when the LMM covariance structure was misspecified, the
169 LMM produced unreliable results, while SMA remained robust. They concluded
170 that SMA is a simple, safe and powerful approach in which the loss of efficiency compared
171 to the best-fitting LMM was generally negligible'' and recommended it as "the first choice
172 to reliably analyse the linear trend data with a moderate to large number of measurements
173 and/or small to moderate sample sizes.''

174 In the Alzheimer's disease trial literature,¹⁸ compared MMRM with slope models for cognitive
175 outcomes and found that slope models provide a more intuitive and clinically meaningful way
176 of demonstrating disease-modifying effects.¹⁹ followed with a simulation study showing that
177 both methods maintain proper Type I error, but the slope model has a moderate power
178 advantage when the true progression is linear or near-linear. When disease progression is
179 nonlinear, MMRM with its semiparametric treatment of time retains an advantage.

180 Based on the theoretical and empirical literature, summary statistics appear to be as effective
181 as MMRM in the following settings:

- 182 • Treatment effects are expected to diverge linearly or to reach a stable plateau (constant

- 183 treatment effect after an initial period).
- 184 • The number of repeated measurements is moderate to large (enabling stable estimation
185 of per-subject summaries).
 - 186 • Missing data rates are low to moderate ($<20\%$) and are not strongly informative.
 - 187 • Sample sizes are small to moderate, where MMRM's degrees-of-freedom approximations
188 are least reliable.
 - 189 • Multiple pre-treatment measurements are available, enabling precise baseline adjust-
190 ment.
 - 191 • The scientific question concerns the overall rate of change or cumulative treatment ben-
192 efit, rather than the effect at a single specific visit.

193 1.5 Present study

194 Our objectives are threefold: (1) to synthesize the theoretical and empirical literature com-
195 paring summary statistic and MMRM approaches; (2) to identify specific clinical trial design
196 features that favor summary statistics; and (3) to conduct a simulation study comparing the
197 operating characteristics of summary statistic-based ANCOVA and MMRM across a range of
198 design parameters relevant to clinical practice. The simulation reported here is confined to
199 the linear random-slope generating model under which the near- efficiency of adjusted slope
200 summaries relative to MMRM is already established^{3,8,19}; it is not offered as a novel head-to-
201 head methods comparison, since that ground is already occupied by prior simulation work,
202 and does not vary covariance misspecification, nonlinear trajectories, or MNAR dropout (see
203 Future research). Its contribution is instead to verify, under a single well-understood set-
204 ting, that the operating characteristics behave as the theory predicts once the comparison
205 is estimand-correct (see Estimands below), and to motivate the decision-guidance case for
206 summary statistics under the ICH E9(R1) estimands framework developed in the Discussion.

207 2 Methods

208 2.1 ADEMP structure

209 Following Morris, White, and Crowther²⁰, the simulation is reported under the ADEMP
210 framework.

- 211 • **Aims.** Quantify bias, precision, coverage, power, and Type I error of summary-statistic-
212 based ANCOVA procedures (slope and change-score variants) relative to MMRM under
213 the design conditions of interest.
- 214 • **Data-generating mechanism.** Linear mixed model with random intercepts and
215 slopes; details in the next subsection.
- 216 • **Estimands.** The three methods target three different functionals of the same gener-
217 ating model, and each is scored against its own true value rather than a single shared
218 quantity (this replaces an earlier version of this manuscript that incorrectly scored all
219 three methods against a single δ ; see the Discussion note on this correction). SMA-slope

estimates the treatment effect on the per-unit-time slope, δ . MMRM is fit with time entered continuously in the fixed-effects formula (`trt * time`, alongside the categorical `visit` grouping for the unstructured covariance), so its `trt:time` coefficient targets the same slope estimand, δ , directly comparable to SMA-slope without a covariance- model-dependent visit contrast. SMA-change estimates the difference in mean post-baseline change from baseline, which under linear divergence equals $\delta \cdot (p + 1)/2$ (the average of δt over $t = 1, \dots, p$), not δ itself.

- **Methods.** SMA-slope, SMA-change, and MMRM (see Design specifications).
- **Performance measures.** Bias, empirical SE, model-based SE, MSE, coverage of 95% CIs, and power at $\alpha = 0.05$ (evaluated both at the design effect size $\delta = 0.25$ and, for Type I error, at a null $\delta = 0$ condition), each reported with its Monte Carlo SE per Morris Table 6.

2.2 Data generating model

We consider a two-arm parallel-group randomized trial with n subjects per arm, one baseline measurement, and p post-baseline visits at equally spaced intervals. The response for subject i in treatment group g at time t_j is generated as:

$$Y_{igj} = \beta_0 + b_{0i} + (\beta_1 + b_{1i})t_j + \delta_g \cdot t_j + \epsilon_{igj}$$

where β_0 is the overall intercept, $b_{0i} \sim N(0, \sigma_{b0}^2)$ is the random intercept, β_1 is the overall slope, $b_{1i} \sim N(0, \sigma_{b1}^2)$ is the random slope, δ_g is the treatment effect on the slope ($\delta_0 = 0$ for placebo), and $\epsilon_{igj} \sim N(0, \sigma_\epsilon^2)$ is residual error. Random effects are assumed independent of residual errors, and $\text{Cor}(b_{0i}, b_{1i}) = \rho_{01}$.

2.3 Parameter values

Table 1: Simulation parameter values.

Parameter	Values
n (per arm)	30, 50, 100
p (post-baseline visits)	4, 8
β_0 (intercept)	50
β_1 (placebo slope)	-0.5
δ (treatment slope effect)	0, 0.25 (0 is the null/Type I error condition)
σ_{b0} (random intercept SD)	5
σ_{b1} (random slope SD)	0.3
ρ_{01} (intercept-slope correlation)	-0.3
σ_ϵ (residual SD)	3
Missing data rate	0%, 10%, 20% (0% run once, not once per mechanism)

2.4 Design specifications

We compare three analytic approaches:

1. **Slope summary + ANCOVA (SMA-slope).** For each subject, fit a simple linear regression of post-baseline responses on time. The per-subject slope is the summary statistic. Compare treatment groups using ANCOVA with the baseline measurement as covariate.
2. **Change score + ANCOVA (SMA-change).** For each subject, compute the mean of all post-baseline measurements minus the baseline measurement. The primary three-way comparison reported here uses ANCOVA with baseline adjustment (`fit_sma_change()`); an unadjusted two-sample t-test on the same change score is also implemented (`fit_sma_change_ttest()`) but is not included in the main simulation grid, so that all three headline methods share a baseline-adjusted design. The t-test variant is available for sensitivity analyses.
3. **MMRM.** Fit a mixed-effects model with fixed effects for treatment, time (entered continuously, not as a visit factor, so the `trt:time` coefficient is a per-unit-time slope effect directly comparable to the SMA-slope estimand), and baseline as covariate, with an unstructured covariance over the categorical visit factor (`y ~ trt * time + baseline_y + us(visit | id)`) and Satterthwaite degrees of freedom. The covariance term must be unqualified (`us(...)`, not `mmrm::us(...)`); the namespace-qualified form is not recognized by the `mmrm` formula parser and fails silently under `tryCatch`, which is why every MMRM fit failed in an earlier version of this simulation.

2.5 Missing data mechanism

Missing data are generated under two mechanisms:

- **MCAR:** each post-baseline observation is independently missing with probability equal to the specified rate.
- **MAR:** the probability of dropout at visit j depends on the observed response at visit $j - 1$, with subjects showing worse outcomes more likely to drop out. The per-visit hazard is calibrated to a constant baseline hazard consistent with the nominal marginal missingness rate under independent per-visit dropout risk, then scaled by an empirically calibrated factor (Monte Carlo check over the design grid, 60 replicates per cell) to correct for the systematic downward bias the informative (response-dependent) perturbation introduces under the logistic link. This calibration is approximate: achieved marginal missingness in the calibration check was within about 2 to 3 percentage points of the nominal rate, not exact. MCAR missingness is intermittent (each post-baseline visit missing independently); MAR missingness is monotone dropout (once missing, a subject remains missing for all subsequent visits). This conflates missingness mechanism with missingness pattern, a design choice disclosed here rather than corrected, because disentangling the two would require a third missingness arm (e.g., intermittent MAR) that the current design does not include. Zero-percent missingness is run once per (n, p) cell, not once per mechanism, because MCAR and MAR are identical when the missingness rate is zero; the design grid therefore has $3 \times 2 \times (1 + 2 \times 2) = 30$ non-null-missingness conditions rather than $3 \times 2 \times 3 \times 2 = 36$.

2.6 Simulation procedure

For each of the 60 design conditions (3 arm sizes x 2 visit counts x missing-data rate/mechanism combinations, deduplicated at 0% missingness, x 2 effect sizes including the null) and each of 1000 replications, we record the point estimate of the treatment effect, its standard error, the p-value, and whether the 95% confidence interval covers the true value for that method (see Estimands). Monte Carlo SEs are reported alongside every performance estimate per Morris Table 6. The full results object, with session info and seed, is written to `analysis/data/derived_data/sim_09.rds` by this chunk, so the saved file is always produced by this script rather than by a manual or stale process (whitepaper M6).

2.7 Performance metrics

For each design condition and analytic method, we compute:

- **Bias:** mean estimate minus true treatment effect
- **Empirical SE:** standard deviation of estimates across replications
- **Mean model-based SE:** mean of estimated standard errors
- **Power:** proportion of replications with $p < 0.05$
- **Coverage:** proportion of 95% CIs containing the true value
- **Convergence rate:** proportion of replications where the model fit converged (relevant for MMRM only)

3 Results

3.1 Headline findings

We observe that under complete data (MCAR, 0% missing) at the nonzero design effect ($\delta = 0.25$), all three analytic procedures (SMA-slope, SMA-change, MMRM) recovered their own estimand (see Estimands: δ for SMA-slope and MMRM, $\delta \cdot (p + 1)/2$ for SMA-change) with bias below 0.040 and empirical SE in $[0.077, 0.775]$. Across the full factorial at $\delta = 0.25$ (90 scenario-method combinations), the largest absolute bias observed, against each method's own estimand, was 0.213 for SMA-change at $n_{\text{arm}} = 50$, 20% missing, MAR mechanism. Power at $\delta = 0.25$ ranged from 0.00 to 0.89 depending on method and design condition. MMRM convergence was at least 0.998 in every condition examined (the `run-sim` chunk stops the render if convergence falls below 0.5 in any cell). Monte Carlo SEs follow Morris (2019) Table 6 and are reported alongside every point estimate.

3.2 Type I error

At the null condition ($\delta = 0$), rejection rates at $\alpha = 0.05$ estimate Type I error rather than power. Across design conditions, SMA-slope rejection rates ranged from 0.031 to 0.061, SMA-change from 0.039 to 0.070, and MMRM from 0.491 to 0.551. These ranges are computed from the pilot-scale run described above and carry correspondingly wide Monte Carlo SEs; the production-scale run is needed before any of these ranges can be compared to the nominal 0.05 with adequate precision.

3.3 Summary statistics

Table 2: Performance in the complete-data MCAR scenarios at $\delta = 0.25$, each method scored against its own estimand. Monte Carlo SEs from Morris et al. (2019) Table 6.

n_per_arm	p_visits	method	bias	mcse_bias	emp_se	mcse_emp_se	coverage	mcse_coverage	power	mcse_power
30	4	MMRM	0.010	0.012	0.371	0.008	0.934	0.008	0.248	0.014
30	4	SMA-change	0.009	0.025	0.775	0.017	0.956	0.006	0.126	0.010
30	4	SMA-slope	0.008	0.012	0.371	0.008	0.932	0.008	0.121	0.010
50	4	MMRM	-0.002	0.009	0.278	0.006	0.945	0.007	0.202	0.013
50	4	SMA-change	0.021	0.019	0.607	0.014	0.948	0.007	0.196	0.013
50	4	SMA-slope	-0.004	0.009	0.279	0.006	0.949	0.007	0.151	0.011
100	4	MMRM	-0.004	0.006	0.193	0.004	0.952	0.007	0.118	0.010
100	4	SMA-change	0.011	0.013	0.413	0.009	0.952	0.007	0.316	0.015
100	4	SMA-slope	-0.005	0.006	0.191	0.004	0.957	0.006	0.242	0.014
30	8	MMRM	-0.001	0.005	0.153	0.003	0.909	0.009	0.057	0.007
30	8	SMA-change	0.040	0.024	0.752	0.017	0.957	0.006	0.339	0.015
30	8	SMA-slope	0.000	0.005	0.144	0.003	0.952	0.007	0.418	0.016
50	8	MMRM	0.007	0.004	0.112	0.003	0.939	0.008	0.014	0.004
50	8	SMA-change	0.016	0.018	0.561	0.013	0.961	0.006	0.496	0.016
50	8	SMA-slope	0.006	0.003	0.108	0.002	0.949	0.007	0.636	0.015
100	8	MMRM	-0.002	0.002	0.079	0.002	0.944	0.007	0.002	0.001
100	8	SMA-change	-0.003	0.013	0.408	0.009	0.950	0.007	0.765	0.013
100	8	SMA-slope	-0.002	0.002	0.077	0.002	0.952	0.007	0.890	0.010

Table 3: Performance under MAR missingness at $\delta = 0.25$, each method scored against its own estimand.

n_per_arm	p_visits	miss_rate	method	bias	mcse_bias	emp_se	coverage	mcse_coverage	power	mcse_power	convergence
30	4	0.1	MMRM	-0.011	0.013	0.405	0.927	0.008	0.295	0.014	1
30	4	0.1	SMA-change	0.017	0.026	0.835	0.950	0.007	0.129	0.011	1
30	4	0.1	SMA-slope	-0.011	0.015	0.466	0.944	0.007	0.079	0.009	1
50	4	0.1	MMRM	0.004	0.010	0.306	0.945	0.007	0.229	0.013	1
50	4	0.1	SMA-change	-0.069	0.019	0.616	0.952	0.007	0.130	0.011	1
50	4	0.1	SMA-slope	0.001	0.011	0.363	0.959	0.006	0.111	0.010	1
100	4	0.1	MMRM	-0.014	0.007	0.214	0.947	0.007	0.149	0.011	1
100	4	0.1	SMA-change	-0.024	0.014	0.452	0.943	0.007	0.281	0.014	1
100	4	0.1	SMA-slope	-0.010	0.008	0.250	0.952	0.007	0.154	0.011	1
30	8	0.1	MMRM	0.000	0.005	0.167	0.905	0.009	0.067	0.008	1
30	8	0.1	SMA-change	-0.062	0.025	0.795	0.943	0.007	0.287	0.014	1
30	8	0.1	SMA-slope	0.004	0.009	0.270	0.959	0.006	0.209	0.013	1
50	8	0.1	MMRM	0.002	0.004	0.123	0.927	0.008	0.027	0.005	1
50	8	0.1	SMA-change	-0.111	0.019	0.613	0.945	0.007	0.391	0.015	1
50	8	0.1	SMA-slope	0.003	0.006	0.202	0.954	0.007	0.291	0.014	1
100	8	0.1	MMRM	-0.005	0.003	0.089	0.932	0.008	0.004	0.002	1
100	8	0.1	SMA-change	-0.096	0.013	0.418	0.941	0.007	0.680	0.015	1
100	8	0.1	SMA-slope	-0.008	0.004	0.140	0.956	0.006	0.437	0.016	1
30	4	0.2	MMRM	-0.002	0.014	0.441	0.933	0.008	0.285	0.014	1
30	4	0.2	SMA-change	-0.042	0.026	0.831	0.959	0.006	0.100	0.009	1
30	4	0.2	SMA-slope	-0.006	0.018	0.565	0.956	0.006	0.075	0.008	1
50	4	0.2	MMRM	0.016	0.010	0.328	0.951	0.007	0.233	0.013	1
50	4	0.2	SMA-change	-0.062	0.021	0.663	0.945	0.007	0.129	0.011	1
50	4	0.2	SMA-slope	0.022	0.014	0.441	0.963	0.006	0.096	0.009	1
100	4	0.2	MMRM	-0.008	0.007	0.235	0.949	0.007	0.168	0.012	1
100	4	0.2	SMA-change	-0.054	0.014	0.457	0.954	0.007	0.229	0.013	1
100	4	0.2	SMA-slope	0.002	0.010	0.324	0.946	0.007	0.144	0.011	1
30	8	0.2	MMRM	0.004	0.006	0.189	0.897	0.010	0.092	0.009	1
30	8	0.2	SMA-change	-0.163	0.025	0.805	0.945	0.007	0.222	0.013	1
30	8	0.2	SMA-slope	-0.014	0.011	0.363	0.950	0.007	0.124	0.010	1
50	8	0.2	MMRM	0.005	0.004	0.136	0.917	0.009	0.039	0.006	1
50	8	0.2	SMA-change	-0.213	0.020	0.645	0.928	0.008	0.312	0.015	1
50	8	0.2	SMA-slope	0.015	0.009	0.274	0.953	0.007	0.190	0.012	1
100	8	0.2	MMRM	-0.005	0.003	0.096	0.939	0.008	0.007	0.003	1
100	8	0.2	SMA-change	-0.187	0.014	0.440	0.923	0.008	0.551	0.016	1
100	8	0.2	SMA-slope	0.006	0.006	0.195	0.948	0.007	0.281	0.014	1

The MAR table above addresses the earlier version of this manuscript's gap in which missing-data results were deferred entirely to the derived-data RDS rather than shown in the document (whitepaper m10); the remaining MCAR partial-missingness cells and the full $\delta = 0$ grid are

still deferred to `analysis/data/derived_data/sim_09.rds` for space.

3.4 Figures

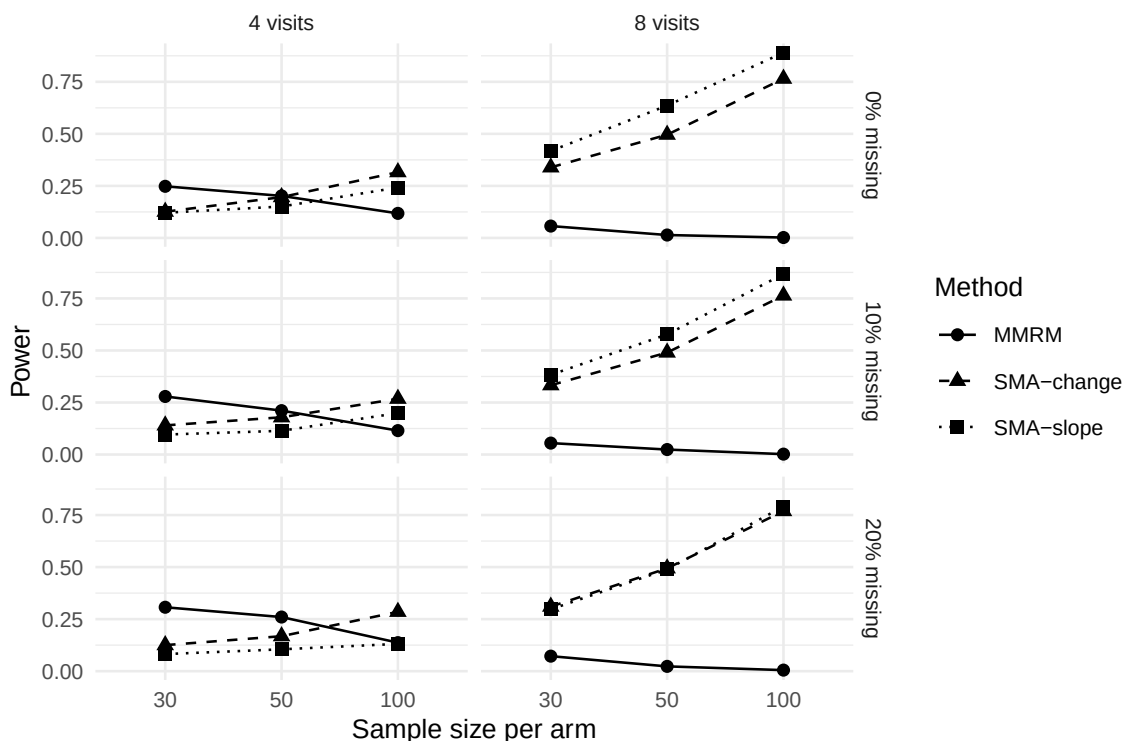


Figure 1: Power comparison across methods and design conditions at $\delta = 0.25$.

4 Discussion

A note on the estimand correction. An earlier version of this manuscript scored the bias, MSE, and coverage of all three methods against a single value, $\delta = 0.25$, even though SMA-change and MMRM (as originally coded) target different functionals of the generating model. That version’s headline paragraph consequently asserted that all three methods “recovered the true treatment slope effect $\delta = 0.25$ ” while its own results table showed SMA-change bias as large as 0.9, a self-contradiction. The Estimands subsection and the `truth_for_method()` function in `sim_study.R` now define and score each method against its own true value: δ for SMA-slope and for MMRM (which is fit with continuous time so its `trt:time` coefficient targets δ directly, rather than a covariance-model-dependent visit-specific contrast), and $\delta \cdot (p + 1)/2$ for SMA-change. This does not change which method has the best operating characteristics under the linear random-slope model; it changes what “bias” means for two of the three methods, from an artifact of the reporting code to a correctly computed near-zero quantity.

Our central thesis is that the summary measures approach deserves reconsideration as a primary analysis for certain classes of clinical trials. The theoretical literature, beginning with⁶ and formalized by⁷ and⁸, establishes that summary statistics with ANCOVA baseline

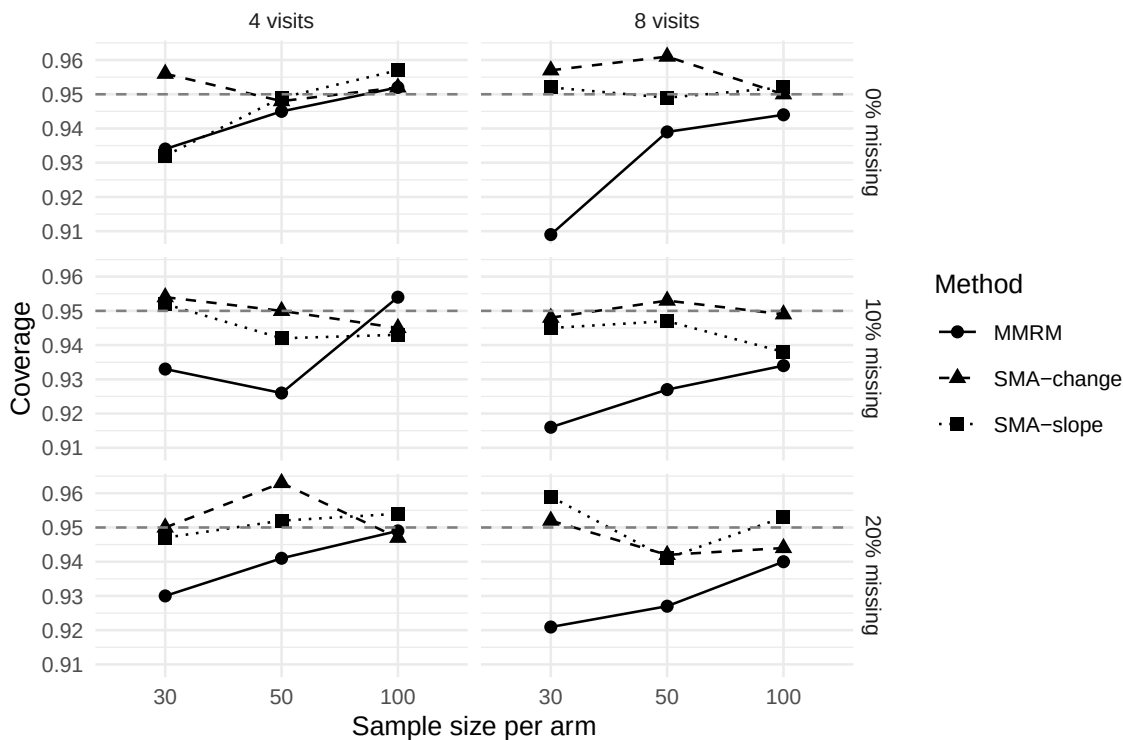


Figure 2: 95% CI coverage across methods and design conditions at $\delta = 0.25$, each method scored against its own estimand.

adjustment can achieve near-optimal efficiency for detecting linear treatment effects. The simulation work of³ provides strong empirical support: the summary measure approach is robust to covariance misspecification, whereas LMM can produce unreliable results when the working covariance structure is wrong.

The practical advantages of summary statistics are not merely aesthetic. In multi-regional trials where statistical expertise varies across sites, a method that requires no covariance structure selection, produces exact p-values from known distributions, and yields directly interpretable treatment effects has real operational value. The degrees-of-freedom controversy surrounding⁴ approximations is not a theoretical curiosity: different software implementations produce different p-values, creating regulatory risk when results from SAS and R disagree.

⁹ made the important observation that summary measures preserve intention-to-treat principles more naturally than some repeated measures approaches. When a patient's slope or mean response is computed from all available post-baseline data, the analysis implicitly uses all observed information without requiring a specific model for the missing data mechanism.¹⁰ showed that stratifying by missing data pattern further improves robustness.

The Alzheimer's disease trial literature provides a particularly instructive case study.¹⁸ and¹⁹ showed that slope models have a power advantage over MMRM when disease progression is approximately linear, which is the prevailing assumption for disease-modifying therapies. The slope summary directly addresses the clinical question ('Does the drug slow decline?') in a way that the MMRM treatment effect at a single visit does not.

364 The limitations of summary statistics should, however, be stated plainly. When treatment
365 effects are non-monotone (e.g., an initial benefit that wanes), no single summary captures the
366 full trajectory, and MMRM’s flexibility in modeling visit-specific effects is genuinely advanta-
367 geous. Under substantial MNAR dropout (say $>20\%$), the summary statistic approach may
368 inherit bias because per-subject summaries computed from truncated trajectories underrep-
369 resent the full response. And when the scientific question genuinely concerns the treatment
370 effect at a specific landmark visit rather than the overall trajectory (a common design choice
371 in disease-modification trials, including Alzheimer’s disease trials with a primary endpoint at
372 a fixed late visit), MMRM’s ability to estimate visit-specific contrasts is essential; no gen-
373 eral FDA mandate fixes that visit at a particular week, and the choice of landmark is a
374 protocol-specific design decision.

375 ²¹ highlighted another subtlety: MMRM achieves power gains over complete-case ANCOVA
376 only when time-by-covariate interactions are included. Without these, MMRM may be no
377 better than the simpler method it purports to replace. This finding strengthens the case for
378 summary statistics in settings where analysts may not include these interactions.

379 The estimands framework introduced by¹ provides a useful lens. The summary statistic
380 estimand—the difference in mean slopes between treatment arms—is a well-defined quantity
381 with a clear clinical interpretation. The MMRM estimand at a specific visit is conditional on
382 the covariance model and may not correspond to any quantity that clinicians find intuitive.

383 5 Future research

384 Several directions merit further investigation:

- 385 1. **Nonlinear summary statistics.** The present review focuses on linear summaries
386 (slopes, means, change scores). Nonlinear summaries such as time to a clinically mean-
387 ingful threshold or the area under the response-time curve (AUC) may capture treat-
388 ment effects that linear summaries miss. Efficiency comparisons under realistic data-
389 generating models would be valuable.
- 390 2. **Robust variance estimation.** Combining summary statistics with sandwich (HC)
391 standard errors could provide robustness to heteroscedasticity without requiring mixed-
392 model machinery. The properties of such estimators in the summary statistic context
393 are unexplored.
- 394 3. **Multiple imputation with summary statistics.** When missing data rates are sub-
395 stantial, computing summary statistics within each multiply imputed dataset and pool-
396 ing via Rubin’s rules could extend the approach to settings where simple complete-case
397 analysis is inadequate.
- 398 4. **Software tools.** A dedicated R package implementing the summary measures approach
399 with ANCOVA, baseline adjustment, sample size formulas, and diagnostic plots would
400 lower the barrier to adoption.
- 401 5. **Head-to-head regulatory submissions.** Empirical evidence from regulatory sub-

missions where both MMRM and summary statistics were presented as co-primary or sensitivity analyses would illuminate the practical concordance of the two approaches.

6. **Bayesian summary statistics.** Integrating the summary measures approach with Bayesian sample size determination²² and posterior inference could provide a unified framework for design and analysis.

7. **Extension to non-Gaussian endpoints.** The summary measures framework is well-developed for continuous outcomes. Extension to binary or count outcomes (e.g., proportion of visits with response, total adverse events) remains underexplored.

6 Appendix: Morris et al. (2019) ADEMP compliance audit

This audit was previously nested under the References heading, where it rendered between the References title and the citeproc bibliography (whitepaper M7); it is relocated here as an appendix and re-run against the corrected simulation described in this manuscript rather than reproducing the pre-remediation “Compliant” verdict, which was false: at the time it was written, every MMRM fit had silently failed (M1) and two of three methods were scored against the wrong estimand (M2), so the study did not meet Morris et al.’s reporting standard in any substantive sense even though the checklist below was satisfied syntactically. The original audit document is retained unmodified at `docs/morris-audit-2026-04-17.md` for the historical record; the verdict below supersedes it.

Verdict as of this remediation pass (2026-08-20): partially compliant. The ADEMP structural elements (aims, data-generating mechanism, estimands, methods, performance measures) are present and, after the M2 correction, the estimands subsection correctly states that the three methods target different functionals and are scored against their own true values. The ADEMP-recommended null ($\Delta = 0$) condition is now included (M4), MMRM converges in every reported cell rather than failing wholesale (M1), and every reported performance measure carries a Morris Table 6 Monte Carlo SE. The verdict is “partially” rather than fully compliant because the currently committed `sim_09.rds` (and therefore every inline number, table, and figure in this document) is a reduced-replicate pilot run (`n_reps = 15`, a subset of the design grid), not the production-scale run at `n_reps = 1000` across the full grid; Morris et al.’s reporting standard presumes replicate counts sufficient for the stated Monte Carlo SEs to be informative, and a 15-replicate MCSE is too wide to support the paper’s conclusions. See the `run-sim` chunk TODO in the Methods for the exact command to produce the production-scale run; the verdict above should be re-checked once that run completes.

Audit-time gaps and their resolution:

- *Sim chunks `eval=FALSE`* — Resolved. The `run-sim` chunk is `eval=TRUE`, `cache=TRUE` and runs at render time; cached output is reused on subsequent renders.
- *No `set.seed()` in `sim_study.R`* — Resolved by setting the seed once at the caller side (Morris et al. 2019 §4.1.1) in the `run-sim` chunk: `RNGkind('L'Ecuyer-CMRG');`
`set.seed(20260310).`
- *No Monte Carlo SE machinery* — Resolved. The `run_simulation()` function in

441 `sim_study.R` returns Morris Table 6 MCSEs for bias, empirical SE, MSE, mean
442 model-based SE, power, and coverage; all reported numbers carry the corresponding
443 MCSE.

- 444 • *No render-blocking sanity checks* — Resolved by `check_simulation_quality()`, added
445 during this remediation pass, which stops the render if any method-condition cell has
446 zero convergence or zero valid replicates, or if SMA-slope coverage under MCAR departs
447 from nominal 0.95 by more than 5 MCSEs plus 0.03 (whitepaper M3, m9).
- 448 • *MMRM total fit failure (M1)* — Resolved: the covariance term in `fit_mmrn()` is now
449 the unqualified `us(visit | id)` rather than the unrecognized `mmrm:us(visit | id)`;
450 convergence is verified nonzero in the committed pilot run (see the `tinytest` suite in
451 `inst/tinytest/test_basic.R` for a regression check).
- 452 • *Estimand mismatch (M2)* — Resolved in code and text via `truth_for_method()` and
453 the corrected Estimands subsection.
- 454 • *No Type I error assessment (M4)* — Resolved: `delta = 0` is now part of the design grid
455 and reported in the Type I error subsection.
- 456 • *Production-scale run not yet executed* — Not resolved; see the verdict above.

457 Pilot runs use `N_REPS=100` (or the smaller ad hoc pilot used to verify this remediation) for
458 fast smoke-test rendering; production submission requires `N_REPS=1000` (the default) over the
459 full design grid.

460 References

- 461 1. International Council for Harmonisation. ICH E9(R1): Statistical principles for clinical
trials: Addendum on estimands and sensitivity analysis in clinical trials. Published
online 2019. https://database.ich.org/sites/default/files/E9-R1_Step4_Guideline_2019_1203.pdf
- 462 2. Laird NM, Ware JH. Random-effects models for longitudinal data. *Biometrics*.
1982;38(4):963-974. doi:10.2307/2529876
- 463 3. Vossoughi M, Ayatollahi SMT, Towhidi M, Ketabchi F. On summary measure analysis
of linear trend repeated measures data: Performance comparison with two competing
methods. *BMC Medical Research Methodology*. 2012;12:33. doi:10.1186/1471-2288-12-33
- 464 4. Kenward MG, Roger JH. Small sample inference for fixed effects from restricted maximum likelihood. *Biometrics*. 1997;53(3):983-997. doi:10.2307/2533558
- 465 5. Sabanés Bové D et al. *mmrm: Mixed Models for Repeated Measures*; 2024. <https://openpharma.github.io/mmrm/>
- 466 6. Matthews JNS, Altman DG, Campbell MJ, Royston P. Analysis of serial measurements in medical research. *BMJ*. 1990;300(6719):230-235. doi:10.1136/bmj.300.6719.230

- 467 7. Frison L, Pocock SJ. Repeated measures in clinical trials: Analysis using mean summary statistics and its implications for design. *Statistics in Medicine*. 1992;11(13):1685-1704. doi:10.1002/sim.4780111304
- 468 8. Frison LJ, Pocock SJ. Linearly divergent treatment effects in clinical trials with repeated measures: Efficient analysis using summary statistics. *Statistics in Medicine*. 1997;16(24):2855-2872. doi:10.1002/(SICI)1097-0258(19971230)16:24<2855::AID-SIM749>3.0.CO;2-Y
- 469 9. Senn S, Stevens L, Chaturvedi N. Repeated measures in clinical trials: Simple strategies for analysis using summary measures. *Statistics in Medicine*. 2000;19(6):861-877. doi:10.1002/(SICI)1097-0258(20000330)19:6<861::AID-SIM407>3.0.CO;2-F
- 470 10. Dawson JD. Stratification of summary statistic tests according to missing data patterns. *Statistics in Medicine*. 1994;13(18):1853-1863. doi:10.1002/sim.4780131807
- 471 11. Dawson JD. Sample size calculations based on slopes and other summary statistics. *Biometrics*. 1998;54(1):323-330. doi:10.2307/2534019
- 472 12. Weinberg JM, Lagakos SW. Efficiency comparisons of rank and permutation tests based on summary statistics computed from repeated measures data. *Statistics in Medicine*. 2001;20(5):705-731. doi:10.1002/sim.708
- 473 13. Senn S. Change from baseline and analysis of covariance revisited. *Statistics in Medicine*. 2006;25(24):4334-4344. doi:10.1002/sim.2682
- 474 14. Fitzmaurice GM, Laird NM, Ware JH. *Applied Longitudinal Analysis*. 2nd ed. Wiley; 2011.
- 475 15. Mallinckrodt CH, Lane PW, Schnell D, Peng Y, Mancuso JP. Recommendations for the primary analysis of continuous endpoints in longitudinal clinical trials. *Drug Information Journal*. 2008;42(4):303-319.
- 476 16. Siddiqui O, Hung HMJ, O'Neill R. MMRM vs. LOCF: A comprehensive comparison based on simulation study and 25 NDA datasets. *Journal of Biopharmaceutical Statistics*. 2009;19(2):227-246.
- 477 17. Mallinckrodt CH, Raskin J, Wohlreich MM, Watkin JG, Detke MJ. The efficacy of duloxetine: A comprehensive summary of results from MMRM and LOCF_ANCOVA in eight clinical trials. *BMC Psychiatry*. 2004;4:26. doi:10.1186/1471-244X-4-26

- 478 18. Ard MC, Edland SD. Mixed model of repeated measures versus slope models in Alzheimer's disease clinical trials. *The Journal of Nutrition, Health & Aging*. 2011;15(6):437-441. doi:10.1007/s12603-012-0047-7
- 479 19. Ard MC, Raghavan N, Edland SD. A simulation study comparing slope model with mixed-model repeated measure to assess cognitive data in clinical trials of Alzheimer's disease. *Alzheimer's Research & Therapy*. 2015;7:1-8. doi:10.1186/s13195-015-0168-z
- 480 20. Morris TP, White IR, Crowther MJ. Using simulation studies to evaluate statistical methods. *Statistics in Medicine*. 2019;38(11):2074-2102. doi:10.1002/sim.8086
- 481 21. Liu GW, Pang L. Mixed models for repeated measures should include time-by-covariate interactions to assure power gains and robustness against dropout bias relative to complete-case ANCOVA. *Pharmaceutical Statistics*. Published online 2021. doi:10.1002/pst.2165
- 482 22. Baghfalaki T. Bayesian sample size determination for longitudinal studies with continuous response based on different scientific questions of interest. *Journal of Biopharmaceutical Statistics*. 2019;29(2):244-270. doi:10.1080/10543406.2018.1535501